

Computer Fundamentals

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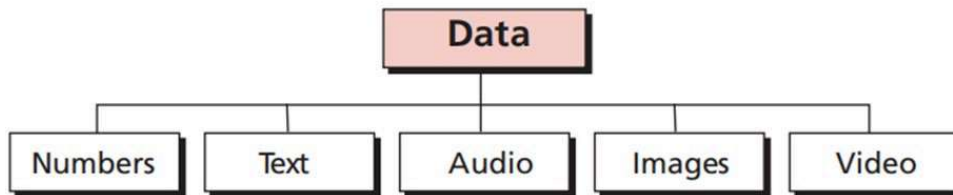
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Chapter2 Data representation and operation

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DATA TYPES

- Data today come in different forms including numbers, text, audio, images, and video.



The computer industry uses the term 'multimedia' to define information that contains numbers, text, audio, images, and video.

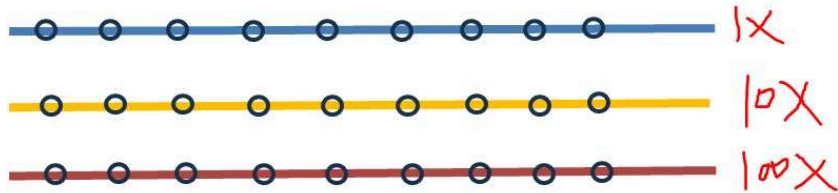
1. Number Systems

- **Number systems**
 - The decimal system (base 10) 十进制
 - The hexadecimal system (base 16)
 - The binary system (base 2)
 - The octal system (base 8)

1. Number Systems

- The Decimal system

- review about decimal



- Symbols(digits): 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
- eg: 18, 1501,
- Digits: the symbols in the system.
- Base: the number of symbols used in the system.

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1. Number Systems

- The ¹⁶Hexadecimal system

- review about hexadecimal

- 半斤八两 half Jin = 8 Liang
- 1 Jin = 16 Liang

- Symbols(digits): 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F
- eg: F, 10F

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1. Number Systems

- **Position number systems:**

System	Base	Symbols(digits)	Eg
Decimal	10	0,1,2,3,4,5,6,7,8,9 [0, b-1]	234.56
Hexadecimal	16	<u>0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F</u>	(AB.1F) ₁₆
Binary	<u>2</u>	0,1	(101.11) ₂
Octal	<u>8</u>	0,1,2,3,4,5,6,7	(163.25) ₈

- $9D + 1D = 10D$, $FH + 1H = 10H$, $(7)8 + (1)8 = (10)8$, $1B + 1B = 10B$
- $393 = \underline{300} + \underline{90} + \underline{3}$
- $\quad = 3 \times 100 + 9 \times 10 + 3 \times 1$
- $\quad = 3 \times 10^2 + 9 \times 10^1 + 3 \times 10^0$ (2,1,0 represent the power of base, and the power corresponds the position of digits)

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1. Number Systems

- **Position number systems(Positional Notation):**

- A system of expressing numbers in which the digits are arranged in succession, the position of each digit has a place value, and the number is equal to the sum of the products of each digit by its place value.
- In a positional number system, the position a symbol occupies in the number determines the value it represents.

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1. Number Systems

Base: number of symbols used in the system. (R=2,8,16,10)

Digits: the symbols in the R system. (R=2,8,16,10)

Place/position of digit: the first digit to the left of the radix point is the 0th place.

Position corresponds the power of digit.

Integer part: the further to the left the digits is, the larger the value.

Fractional part: the further to the right the digit is, the smaller the value.

Place value: different position determines the different value.

1. Number Systems

eg: $393.124 = 3 \cdot 10^2 + 9 \cdot 10^1 + 3 \cdot 10^0 + 1 \cdot 10^{-1} + 2 \cdot 10^{-2} + 4 \cdot 10^{-3}$

digit	3	9	3	1	2	4
position	2	1	0	-1	-2	-3
power	2	1	0	-1	-2	-3
weight	10^2	10^1	10^0	10^{-1}	10^{-2}	10^{-3}

1. Number Systems

- **Convert the base R to base 10:**
 - Determine the value of a base R system number
 - Eg: base 16 number (19A.D)H
 - Eg: 14H=(?)D
 - Eg: (257)₈=(?)D
 - Eg: 101011.101B=(?)D

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2. Conversion

- **Summary of the four positional systems**
 - summary and examples

System	Base	Symbols	Examples
Decimal	10	0, 1, 2, 3, 4, 5, 6, 7, 8, 9	2345.56
Binary	2	0, 1	(1001.11) ₂
Octal	8	0, 1, 2, 3, 4, 5, 6, 7	(156.23) ₈
Hexadecimal	16	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F	(A2C.A1) ₁₆

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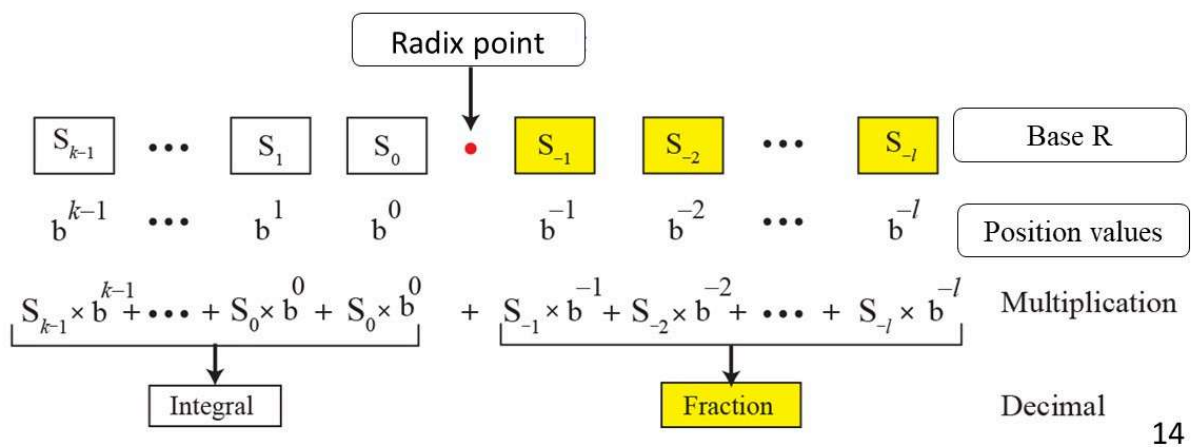
2. Conversion

- **Summary of the four positional systems**
 - comparison of numbers in the four systems

Decimal	Binary	Octal	Hexadecimal
0	0	0	0
1	1	1	1
2	10	2	2
3	11	3	3
4	100	4	4
5	101	5	5
6	110	6	6
7	111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	13 F

2. Conversion

- **Any base to decimal conversion**
 - solution



2. Conversion

- Any base to decimal conversion
 - examples

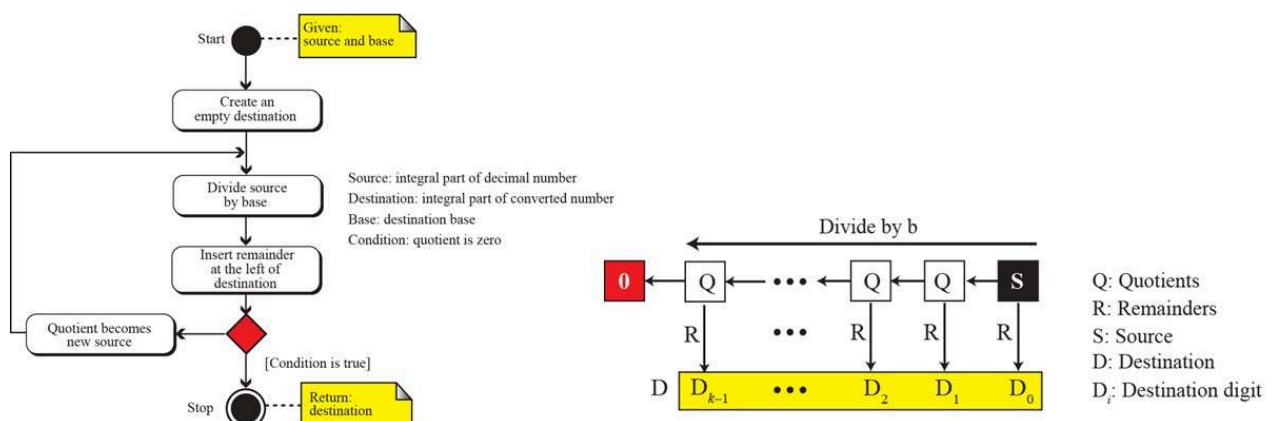
19A.DH=(?)D

1011.101B=(?)D

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2. Conversion

- Decimal to any base
 - (1) Converting the integral part



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2. Conversion

- **Converting the integral part**
 - examples(binary): addition method

position	7	6	5	4	3	2	1	0	-1	-2	-3
	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	2^{-1}	2^{-2}	2^{-3}
value	128	64	32	16	8	4	2	1	0.5	0.25	0.125

9=(?)B

98=(?)B

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2. Conversion

- **Converting the integral part**
 - examples (binary): division method

98=(1100010)B

2	98		remainder	
2	49	-----	0	↑ Low position
2	24	-----	1	
2	12	-----	0	
2	6	-----	0	
2	3	-----	0	
2	1	-----	1	
	0	-----	1	High position

$$\begin{aligned}
 98 &= 2 * 49 + 0 \\
 &= 2 * (2 * 24 + 1) + 0 \\
 &= 2 * (2 * (2 * 12 + 0) + 1) + 0 \\
 &= 2 * (2 * (2 * (2 * 6 + 0) + 0) + 1) + 0 \\
 &= 2 * (2 * (2 * (2 * (2 * 3 + 0) + 0) + 0) + 1) + 0 \\
 &= 2 * (2 * (2 * (2 * (2 * 1 + 1) + 0) + 0) + 0) + 1) + 0
 \end{aligned}$$

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2. Conversion

- **Converting the integral part**
 - Examples(hexadecimal)

479=(?)H

254=(?)H

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2. Conversion

- **Converting the integral part**
 - Examples(octal)

25=(?)8

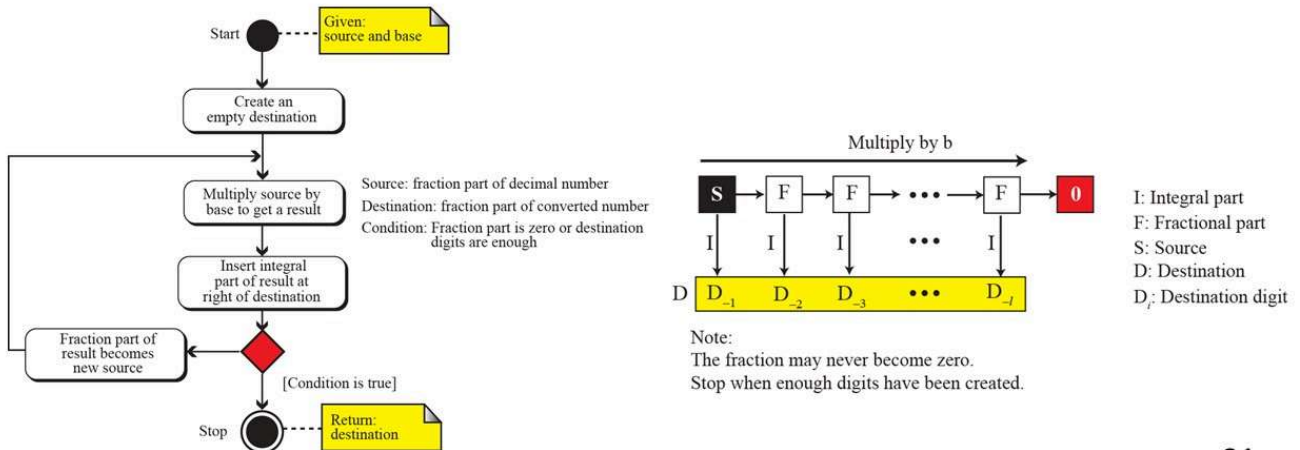
112=(?)8

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2. Conversion

- **Decimal to any base**

- **(2) Converting the fractional part**



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2. Conversion

- **Converting the fractional part**

- **examples (binary): multiplication method**

$$0.125 = (?)_B$$

$$0.2 = (?)_B$$

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2. Conversion

- **Converting the fractional part**
 - examples (hexadecimal): multiplication method

$0.625 = (?)_H$

$0.4 = (?)_H$

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2. Conversion

- **Converting the fractional part**
 - examples (octal): multiplication method

$0.625 = (?)_8$

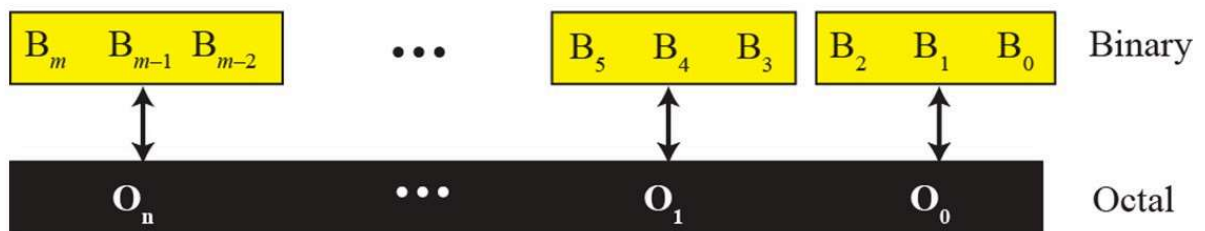
$0.4 = (?)_8$

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2. Conversion

- **binary-octal conversion**
 - three bits is one octal digit.

B_i : Binary digit (bit) O_i : Octal digit



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2. Conversion

- **binary-octal conversion**
 - three bits is one octal digit.

octal	binary
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

10110101.101010B=()8

1111.11 =()8

(123.25)8=()B

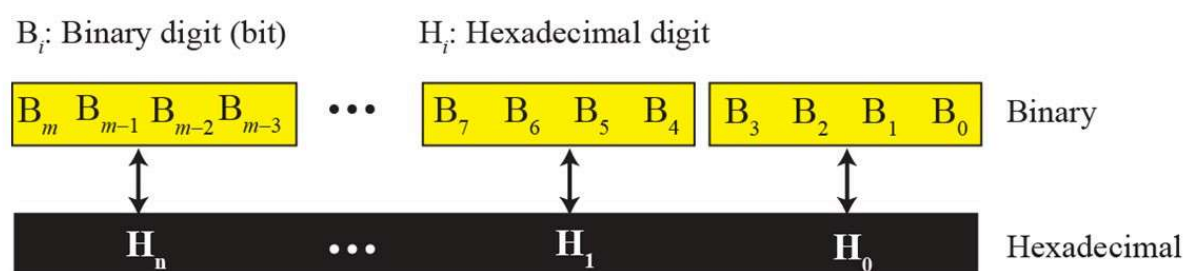
(26.32)8 =()B

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2. Conversion

- binary-hexadecimal conversion**

- four bits is one hexadecimal digit.



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2. Conversion

- binary-hexadecimal conversion**

- four bits is one hexadecimal digit.

10110101.10101000B=()H

11001.101B=()H

38.D2H=()B

16.EH=()B

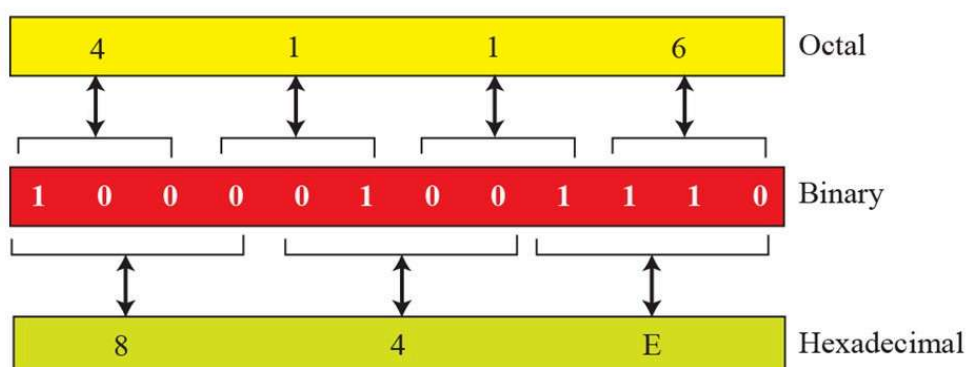
hexadecimal	binary
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
A	1010
B	1011
C	1100
D	1101
E	1110
F	1111

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2. Conversion

- **octal-hexadecimal conversion**

- We first convert the number to binary, then we rearrange the bits in groups of three/four bits to find the octal/hexadecimal equivalent.



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2. Conversion

- **octal-hexadecimal conversion**

$$(4116.1)_8 = (\quad)_{16}$$

$$\begin{aligned} (4116.1)_8 &= (100\ 001\ 001\ 110.001)_B \\ &= (1000\ 0100\ 1110.0010)_B \\ &= (8\ 4\ E\ .2)_{16} \\ &= (84E.2)_{16} \end{aligned}$$

Decimal	Binary	Octal	Hexadecimal
0	0	0	0
1	1	1	1
2	10	2	2
3	11	3	3
4	100	4	4
5	101	5	5
6	110	6	6
7	111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

单选题 1分

1. Which number shouldn't be octal.

- ☐ A 123
- ☐ B 1011.101
- ☐ C 28.6
- ☐ D 642

填空题 1分

2. Convert the number 110D to binary [填空1]

填空题 1分

3. Convert the number 12.625D to binary [填空1]

填空题 1分

4. Convert the number 14.75D to octal [填空1]

填空题 1分

5. Convert the number 21.375D to hexadecimal [填空1]

填空题 1分

6. Convert the number 11001.110B to hexadecimal [填空1]

填空题 1分

7. Convert the number 11001.110B to octal [填空1]

填空题 1分

8. Convert the number A8.1H to octal [填空1]